Project Report

DATA ANALYTICS DASHBOARD USING STREAMLIT

Machine Learning Based-Employee Attrition Prediction and Risk Scoring System

Submitted By

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Technology Used

Programming language:

- Python

Libraries:

- Pandas
- Scikit-learn
- NumPy
- Plotly Express
- Streamlit
- Sklearn

Machine Learning Algorithm:

- Random Forest Classifier

Project Domain

HR Analytics

Submitted to

Unified mentor private limited

Abstract

Employee attrition is a significant challenge for organizations as it affects productivity, increases recruitment costs, and impacts overall business performance. Predicting employee attrition in advance enables Human Resource (HR) departments to take proactive measures to improve employee retention. This project, titled **"Employee Attrition Prediction & Risk Scoring System,"** aims to analyze employee data and identify employees who are at risk of leaving the organization using machine learning techniques.

The system is developed using **Python**, **Pandas**, **Scikit-learn**, **Plotly**, and **Streamlit**. A **Random Forest Classifier** is used to predict employee attrition based on various factors such as department, monthly income, job satisfaction, work-life balance, years at the company, and other employee attributes. The processed data is presented through an interactive Streamlit dashboard that provides key insights, including employee statistics, attrition distribution, department-wise analysis, risk categorization, and model performance metrics.

The dashboard enables HR professionals to filter employee information, visualize trends, and monitor attrition rates efficiently. The machine learning model achieved satisfactory performance, demonstrating its capability to identify employees with a higher likelihood of attrition. This project highlights the practical application of data analytics and machine learning in HR management and serves as a decision-support tool for improving employee retention strategies. Future enhancements may include real-time prediction, explainable AI techniques, and integration with live HR databases.

Introduction

Employee attrition is one of the major challenges faced by organizations. High employee turnover leads to increased recruitment costs, reduced productivity, and loss of experienced talent. Predicting employee attrition helps Human Resource (HR) departments identify employees who are at risk of leaving and take preventive measures.

This project develops an Employee Attrition Prediction and Risk Scoring System using Machine Learning and Streamlit. The system analyzes employee information, predicts attrition, and presents the results through an interactive dashboard.

Problem Statement

However, HR leaders still lack the ability to answer:
Which specific employees are most likely to leave in the near future?
This project introduces predictive intelligence that enables:

- Early identification of high-risk employees.
- Targeted retention interventions.
- Targeted retention interventions.
- Data-driven workforce planning.

Problem Statement

Palo Alto Networks Faces Challenges Such as:

- Sudden, unanticipated resignations.
- Loss of high-performing or critical employees.
- Reactive counter-offers that come too late.

The organization lacks:

- Support operational cost optimization.
- Improve passenger comfort and safety.
- Enable strategic planning for seasonal operations.

Dataset Description

Field Name	Description
Age	Age of the employee in years
Attrition	Indicates whether the employee left the organization (0 / 1)
Business Travel	Frequency of business travel (Non-Travel, Travel Rarely, Travel Frequently)
Daily Rate	Daily compensation rate of the employee
Department	Department where the employee works (e.g., R&D, Sales, HR)
Distance From Home	Distance between employee's home and workplace
Education	Education level (1:Below College, 2:College, 3:Bachelor, 4:Master, 5:Doctor)
Education Field	Field of education (e.g., Life Sciences, Medical, Technical)
Environment Satisfaction	Satisfaction with the work environment (1 = Low, 4 = High)
Gender	Gender of the employee (Male / Female)
Hourly Rate	Hourly wage of the employee
Job involvement	Level of involvement in work (1 = Low, 4 = High)
Job Role	Specific job role or designation of the employee
Job Satisfaction	Satisfaction with the job role (1 = Low, 4 = High)
Marital Status	Marital status of the employee (Single, Married, Divorced)
Monthly income	Monthly salary earned by the employee
Monthly Rate	Monthly compensation rate (administrative pay metric)
Num Companies Worked	Number of companies the employee has worked for previously
Over Time	Indicates whether the employee works overtime (Yes / No)
Percent Salary Hike	Percentage increase in salary during the last appraisal
Performance Rating	Performance evaluation score of the employee

Field Name	Description
Relationship Satisfaction	Satisfaction with workplace relationships (1 = Low, 4 = High)
Stock Optional Level	Level of stock options granted to the employee
Total Working Years	Total professional experience across all companies (in years)
Training Times Last year	Number of training programms attended in the last year
Work Life Balance	Work-life balance rating (1 = Poor, 4 = Excellent)
Years At Company	Number of years the employee has worked at Palo Alto Networks
Years In Current Role	Number of years in the current role
Years Since Last Promotion	Number of years since the last promotion
Years With Curr Manager	Number of years working with the current manager

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Data Science Methodology

Data Preprocessing

- Encode categorical variables(Label/One-Hot Encoding)
- Scale numerical features
- Handle class imbalance(SMOTE or class weights)
- Train-test split with stratification

Feature Engineering

- Income-to-experience ratio
- Promotion delay indicators
- Engagement composite score
- Workload stress flag

Model Development

- **Baseline Models**

Logistic Regression(interpretability baseline)

- **Advanced Models**

Random Forest Classifier

Gradient Boosting/ XGBoost

Model Evaluation

METRIC	**PURPOSE**
Accuracy	Overall correctness
Precision	Overall correctness
Recall	Attrition detection capability
F1-score	Balance between precision & recall
ROC-AUC	Overall classification power

Risk Scoring Framework

Each employee is assigned:

- Attrition Probability(0-1)
- Risk Category:
  Low Risk(<30%)
  Medium Risk(30-60%)
  High Risk (>60%)

Model Explainability

To ensure HR trust:

- Feature importance analysis
- SHAP value explanations(optional)
- Individual-level reason codes(e.g., low satisfaction + high overtime)

Dashboard Features

Executive Dashboard

Displays:

- Total Employees
- Employees Left
- Attrition Rate
- Average Monthly Income

Attrition Distribution

Pie chart showing:

- Employees who stayed
- Employees who left

Employee Data

Displays employee records in tabular format.

Risk Analysis

Employees are categorized into:

- Low Risk
- High Risk

Using attrition labels.

Department Analysis

Shows:

- Employee count by department
- Average monthly income by department

Model Performance

Displays:

- Accuracy: 84.01%
- Precision: 50%
- Recall: 27.66%
- ROC-AUC: 79.40%

Results

The dashboard successfully provides:

- Interactive visualization of employee data.
- Department-wise employee analysis.
- Employee attrition distribution.
- Risk categorization.
- Machine learning model evaluation.

The Random Forest model achieved good accuracy while demonstrating the ability to identify employee attrition patterns.

Advantages

- Easy to use.
- Interactive dashboard.
- Supports HR decision-making.
- Reduces employee turnover risk.
- Visual representation of employee statistics.
- Beginner-friendly implementation.

Limitations

- Uses a single dataset.
- Risk categories are based on attrition labels instead of prediction probabilities.
- Recall can be improved through model tuning.
- Does not provide real-time predictions.

Future Enhancements

- Real-time employee prediction form.
- Feature Importance visualization.
- SHAP Explain ability.
- Confusion Matrix visualization.
- Download reports in PDF or Excel.
- Integration with company HR databases.
- Hyper parameter tuning to improve model performance.

SOURCE CODE SNAPSHOT

```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()

X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
lr = LogisticRegression(
    class_weight='balanced',
    max_iter=5000,
    solver='lbfgs',
    random_state=42
)

lr.fit(X_train_scaled, y_train)


from sklearn.ensemble import GradientBoostingClassifier

gb = GradientBoostingClassifier(
    random_state=42
)

gb.fit(X_train, y_train)

y_pred_gb = gb.predict(X_test)
y_prob_gb = gb.predict_proba(X_test)[:,1]
```

CONCLUSION

The Employee Attrition Prediction & Risk Scoring System demonstrates how machine learning can support HR analytics by identifying employees who are at risk of leaving an organization. The Streamlit dashboard provides an interactive platform for visualizing employee information, analyzing department-wise trends, and evaluating model performance. This project serves as a practical application of data analytics and machine learning concepts and can be extended with advanced predictive techniques for real-world HR management.